# Venkata Lakshmi Parimala Pasupuleti

Software Engineer · AI/ML · Python · ML Infrastructure · Model Evaluation · GenAI & LLM Systems · Data Processing · Quantitative Analysis · Production Debugging

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## PROFESSIONAL SUMMARY

Machine learning engineer and software developer with 8 years of production Python development and two AI systems that demonstrate ML infrastructure, model evaluation, and GenAI/LLM in practice. AgentProbe is a CI-integrated LLM evaluation platform built end-to-end: 15-endpoint FastAPI service, 5-metric quality engine (correctness, groundedness, latency, token efficiency, hallucination rate), model optimization feedback loops, multi-provider routing - zero regressions shipped across 1,240+ test cases. Leasel is an 8-agent agentic AI platform (LangGraph, GenAI/LLM, tool-calling, structured outputs, HITL) using Claude Code as a daily accelerator; 93.6% precision across 1,240 compliance test cases. At Entrata, owned data processing pipelines serving 100K+ housing units - debugging, triage, query optimization, and design reviews built into every sprint across 120+ Agile cycles. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, quantitative methods and ML modeling, May 2026).

## TECHNICAL SKILLS

**ML Infrastructure & Systems:** Python (8yr), ML Infrastructure, Model Evaluation, Model Optimization, Model Deployment, Data Processing Pipelines, FastAPI, REST APIs, PostgreSQL, MySQL, Oracle, Redis, AWS (S3, EC2, RDS, Glue), Docker, CI/CD (GitHub Actions), RabbitMQ, Microservices, Linux, Git

**GenAI & LLM Systems:** Claude Code, GenAI/LLM, LangGraph, LangChain, Agentic AI, Multi-Agent Orchestration, Tool Calling, Structured Outputs, LLM Evaluation, LLM-as-Judge, Hallucination Detection, Optimization Algorithms, RAG Pipelines, ChromaDB, Embeddings, Prompt Engineering, Anthropic API, OpenAI API, Ollama, MCP, HITL, Guardrails

**ML Modeling & Quantitative:** ML Modeling, Data Structures & Algorithms, Quantitative Analysis, Statistical Modeling, Feature Engineering, Isolation Forest, Random Forest, Logistic Regression, Anomaly Detection, PyTorch, scikit-learn, spaCy, BERT, NLP, Debugging & Triage, Design Reviews & Code Reviews, System Design, TypeScript, JavaScript, React.js, Agile/Scrum

## EXPERIENCE

### Magizhs Tech Solution

May 2025 - Aug 2025

**AI/ML Engineer Intern**

- Built a production agentic AI system (LangGraph, FastAPI, Python) from ambiguous requirements through deployment - used Claude Code throughout for rapid prototyping and debugging; owned ML infrastructure setup, testing, and stakeholder handoff end-to-end.

### George Mason University

Aug 2024 - May 2026

**Graduate Teaching Assistant & Research Associate**

- Built ML modeling pipelines (Python, PyTorch, scikit-learn) on WHO and UK Road Safety datasets - $R^2 = 0.977$ regression accuracy; full data processing pipeline from ingestion through feature engineering, quantitative analysis, and statistical validation across 190+ countries.
- Applied Isolation Forest anomaly detection across 3 research pipelines - automated data integrity checks and debugging that surfaced anomalies missed by manual review; designed observability dashboards improving research comprehension 40%.

### Entrata - PropTech / SaaS (100K+ MAU)

Sep 2018 - Aug 2024

**Senior Software Engineer · Lehi, UT (Remote)**

- **Data processing pipelines and ML infrastructure at production scale** - owned HUD income-limit data processing pipeline for 7 years (federal source data → 180,000 domain records powering eligibility and rent calculations for 100K+ housing units); RabbitMQ async processing with retry logic, idempotency guarantees, and dead-letter handling; 200+ SQL query optimizations (30% performance gain via execution plan analysis, indexing, and partitioning); Docker containerization (50% scalability improvement); AWS S3, EC2, RDS, Glue; 99.5% uptime.
- **Production debugging, triage, and optimization** - maintained sub-10 bug count for 6+ consecutive months across 20+ production modules; built internal debugging tools reducing triage time by 40%; 100% PHI compliance with Libsodium-encrypted data pipelines (zero security incidents across 7 years); owned issues end-to-end from source analysis through resolution and post-fix validation.
- **Design reviews, code quality, and cross-functional delivery** - 300+ peer design and code reviews enforcing data structures, algorithms, testability, and performance standards; leveraged Claude Code and AI development tools to accelerate debugging and iteration; mentored 5+ engineers; shipped 20+ production modules across 120+ Agile sprints with product, compliance, and operations stakeholders.

### Entrata India - Pune

Jan 2017 - Sep 2018

**Associate Software Engineer → Trainee**

- Built backend modules (PHP, Python, PostgreSQL) for data processing and compliance workflows; data pipeline for 1.6M HUD/Tax Credit records; AWS S3 migration resolving 90% of legacy storage constraints.

## PROJECTS

### AgentProbe: ML Evaluation Infrastructure with CI Regression Gate

ML Infrastructure · Model Evaluation · GenAI/LLM

Python · FastAPI · LangGraph · ChromaDB · CI/CD (GitHub Actions) · Anthropic API · OpenAI · Ollama | github.com/venkatalakshmiparimala/AgentProbe

Production ML evaluation platform built end-to-end: 15-endpoint FastAPI REST service, 5-metric quality engine (correctness, groundedness, latency, token efficiency, hallucination rate), model optimization feedback loops, CI-integrated regression gate, multi-provider routing for cost optimization. Implemented GenAI/LLM evaluation infrastructure with LLM-as-Judge scoring, structured output validation, and debugging tooling - zero regressions shipped across 1,240+ test cases. Open-source.

### Leasel: 8-Agent GenAI/LLM Platform - Concept Through Production

GenAI · LLM · Agentic AI

Python · LangGraph (8-agent) · FastAPI · TF-IDF · OCR · SQLite · Claude Code | github.com/venkatalakshmiparimala/Lease_complaince_rag

Three-surface GenAI/LLM platform built solo with Claude Code: document intelligence (OCR, section-aware chunking, citation verification), 8-agent LangGraph orchestration (tool routing, structured outputs, HITL escalation), and tenant self-service with deterministic guardrails. Full ML modeling and evaluation harness - 93.6% precision, 87.9% F1 across 1,240 compliance test cases; pass@k with LLM-as-Judge scoring.

### CrashClarity: Quantitative ML on 104K Road Accident Records

ML Modeling · Quantitative Analysis · Data Processing

Python · R · scikit-learn · Random Forest · Logistic Regression · Feature Engineering · GitHub Pages | Portfolio

End-to-end quantitative ML pipeline on 104K+ UK accident records - full data processing from ingestion through feature engineering across weather, road, and temporal variables; Random Forest + Logistic Regression with 85%+ accuracy; data structures and algorithms for efficient record processing; interactive observability dashboard deployed on GitHub Pages for model debugging and result validation.

## EDUCATION

**M.S. Data Analytics Engineering -** George Mason University, Virginia — May 2026 · GPA 3.87

**B.Tech Computer Science & Engineering -** JNTU Anantapur — 2013-2017